

ASSIGNMENT - 3

Master Node

```
2026-02-25 01:14:58,058 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting namenodes on [localhost]
Starting datanodes
10.32.161.43: budda_pallavi1@10.32.161.43: Permission denied (publickey,password).
172.22.109.86: ssh: connect to host 172.22.109.86 port 22: Connection timed out
Starting secondary namenodes [buddapallavi1]
2026-02-25 01:15:14,316 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
budda_pallavi1@buddapallavi1:~/hadoop-3.4.2$ $USER
budda_pallavi1: command not found
budda_pallavi1@buddapallavi1:~/hadoop-3.4.2$ echo $USER
budda_pallavi1
budda_pallavi1@buddapallavi1:~/hadoop-3.4.2$ ssh-copy-id budda_pallavi1@10.32.161.43
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
budda_pallavi1@10.32.161.43's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'budda_pallavi1@10.32.161.43'"
and check to make sure that only the key(s) you wanted were added.

budda_pallavi1@buddapallavi1:~/hadoop-3.4.2$ sbin/start-dfs.sh
Starting namenodes on [localhost]
localhost: namenode is running as process 11014. Stop it first and ensure /tmp/hadoop-budda_pallavi1-namenode.pid file is empty before retry.
Starting datanodes
10.32.161.43: bash: line 1: /home/budda_pallavi1/hadoop-3.4.2/bin/hdfs: No such file or directory
172.22.109.86: ssh: connect to host 172.22.109.86 port 22: Connection timed out
Starting secondary namenodes [buddapallavi1]
buddapallavi1: secondarynamenode is running as process 11288. Stop it first and ensure /tmp/hadoop-budda_pallavi1-secondarynamenode.pid file is empty before retry.
2026-02-25 01:21:08,468 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
budda_pallavi1@buddapallavi1:~/hadoop-3.4.2$
```

```
Feb 25 01:14
samyuktha_new@samyukthanew: ~
budda_pallavi@buddapallavi1:~/hadoop-3.4.2$ ssh samyuktha_new@10.32.161.43
samyuktha_new@10.32.161.43's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-101-generic aarch64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Feb 24 07:43:35 PM UTC 2026

System load:          0.01
Usage of /:           87.6% of 29.82GB
Memory usage:         14%
Swap usage:           0%
Processes:            214
Users logged in:      1
IPv4 address for enp0s1: 10.32.161.43
IPv6 address for enp0s1: 2409:40c1:1029:57d9:e34c:67f1:c4f0:f7a9
IPv6 address for enp0s1: 2409:40c1:1029:57d9:9415:94ff:fec5:c562

=> / is using 87.6% of 29.82GB

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

160 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Nov  6 11:15:50 2025 from 127.0.0.1
samyuktha_new@samyukthanew: $
```

Data Node 1

```
(spark_env) user@DESKTOP-00TJMH0:~$
(spark_env) user@DESKTOP-00TJMH0:~$ sudo systemctl start ssh
(spark_env) user@DESKTOP-00TJMH0:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.
(spark_env) user@DESKTOP-00TJMH0:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-02-24 19:28:50 UTC; 24s ago
 TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Main PID: 2197 (sshd)
     Tasks: 1 (limit: 9344)
    Memory: 1.2M (peak: 1.6M)
       CPU: 22ms
    CGroup: /system.slice/ssh.service
            └─2197 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 24 19:28:50 DESKTOP-00TJMH0 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 24 19:28:50 DESKTOP-00TJMH0 sshd[2197]: Server listening on 0.0.0.0 port 22.
Feb 24 19:28:50 DESKTOP-00TJMH0 sshd[2197]: Server listening on :: port 22.
Feb 24 19:28:50 DESKTOP-00TJMH0 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
(spark_env) user@DESKTOP-00TJMH0:~$
```

```

(spark_env) user@DESKTOP-00TJMH0:~$ cd ~/spark
(spark_env) user@DESKTOP-00TJMH0:~/spark$ sbin/start-worker.sh spark://10.32.161.105:7077
starting org.apache.spark.deploy.worker.Worker, logging to /home/user/spark/logs/spark-user-org.apache.spark.deploy.worker.Worker-1-DESKTOP-00TJMH0.out
(spark_env) user@DESKTOP-00TJMH0:~/spark$ jps
544 Jps
457 Worker
(spark_env) user@DESKTOP-00TJMH0:~/spark$ tail -n 50 logs/spark-user-org.apache.spark.deploy.worker.Worker-1-*.out
at org.apache.spark.network.util.TransportFrameDecoder.channelRead(TransportFrameDecoder.java:102)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:444)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:420)
at io.netty.channel.AbstractChannelHandlerContext.fireChannelRead(AbstractChannelHandlerContext.java:412)
at io.netty.channel.DefaultChannelPipeline$HeadContext.channelRead(DefaultChannelPipeline.java:1410)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:440)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:420)
at io.netty.channel.DefaultChannelPipeline.fireChannelRead(DefaultChannelPipeline.java:919)
at io.netty.channel.nio.AbstractNioByteChannel$NioByteUnsafe.read(AbstractNioByteChannel.java:166)
at io.netty.channel.nio.NioEventLoop.processSelectedKey(NioEventLoop.java:788)
at io.netty.channel.nio.NioEventLoop.processSelectedKeysOptimized(NioEventLoop.java:724)
at io.netty.channel.nio.NioEventLoop.processSelectedKeys(NioEventLoop.java:650)
at io.netty.channel.nio.NioEventLoop.run(NioEventLoop.java:562)
at io.netty.util.concurrent.SingleThreadEventExecutor$4.run(SingleThreadEventExecutor.java:997)
at io.netty.util.internal.ThreadExecutorMap$2.run(ThreadExecutorMap.java:74)
at io.netty.util.concurrent.FastThreadLocalRunnable.run(FastThreadLocalRunnable.java:30)
at java.base/java.lang.Thread.run(Thread.java:840)

```

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at io.netty.util.internal.ThreadExecutorMap$2.run(ThreadExecutorMap.java:74)
at io.netty.util.concurrent.FastThreadLocalRunnable.run(FastThreadLocalRunnable.java:30)
at java.base/java.lang.Thread.run(Thread.java:840)

at org.apache.spark.network.client.TransportResponseHandler.handle(TransportResponseHandler.java:209)
at org.apache.spark.network.server.TransportChannelHandler.channelRead0(TransportChannelHandler.java:142)
at org.apache.spark.network.server.TransportChannelHandler.channelRead0(TransportChannelHandler.java:53)
at io.netty.channel.SimpleChannelInboundHandler.channelRead(SimpleChannelInboundHandler.java:99)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:444)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:420)
at io.netty.channel.AbstractChannelHandlerContext.fireChannelRead(AbstractChannelHandlerContext.java:412)
at io.netty.handler.timeout.IdleStateHandler.channelRead(IdleStateHandler.java:286)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:442)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:420)
at io.netty.channel.AbstractChannelHandlerContext.fireChannelRead(AbstractChannelHandlerContext.java:412)
at io.netty.handler.codec.MessageToMessageDecoder.channelRead(MessageToMessageDecoder.java:103)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:444)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:420)
at io.netty.channel.AbstractChannelHandlerContext.fireChannelRead(AbstractChannelHandlerContext.java:412)
at org.apache.spark.network.util.TransportFrameDecoder.channelRead(TransportFrameDecoder.java:102)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:444)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:420)
at io.netty.channel.AbstractChannelHandlerContext.fireChannelRead(AbstractChannelHandlerContext.java:412)
at io.netty.channel.DefaultChannelPipeline$HeadContext.channelRead(DefaultChannelPipeline.java:1410)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:440)
at io.netty.channel.AbstractChannelHandlerContext.invokeChannelRead(AbstractChannelHandlerContext.java:420)
at io.netty.channel.DefaultChannelPipeline.fireChannelRead(DefaultChannelPipeline.java:919)
at io.netty.channel.nio.AbstractNioByteChannel$NioByteUnsafe.read(AbstractNioByteChannel.java:166)
at io.netty.channel.nio.NioEventLoop.processSelectedKey(NioEventLoop.java:788)
at io.netty.channel.nio.NioEventLoop.processSelectedKeysOptimized(NioEventLoop.java:724)

```



```

    at io.netty.util.concurrent.SingleThreadEventExecutor$4.run(SingleThreadEventExecutor.java:997)
    at io.netty.util.internal.ThreadExecutorMap$2.run(ThreadExecutorMap.java:74)
    at io.netty.util.concurrent.FastThreadLocalRunnable.run(FastThreadLocalRunnable.java:30)
    ... 1 more
(spark_env) user@DESKTOP-00TJMH0:~/spark$ ping 10.32.161.105
PING 10.32.161.105 (10.32.161.105) 56(84) bytes of data.
64 bytes from 10.32.161.105: icmp_seq=1 ttl=63 time=9.55 ms
64 bytes from 10.32.161.105: icmp_seq=2 ttl=63 time=46.2 ms
64 bytes from 10.32.161.105: icmp_seq=3 ttl=63 time=6.47 ms
64 bytes from 10.32.161.105: icmp_seq=4 ttl=63 time=70.2 ms
64 bytes from 10.32.161.105: icmp_seq=5 ttl=63 time=49.3 ms
64 bytes from 10.32.161.105: icmp_seq=6 ttl=63 time=15.4 ms
64 bytes from 10.32.161.105: icmp_seq=7 ttl=63 time=30.5 ms
64 bytes from 10.32.161.105: icmp_seq=8 ttl=63 time=156 ms
64 bytes from 10.32.161.105: icmp_seq=9 ttl=63 time=45.1 ms
64 bytes from 10.32.161.105: icmp_seq=10 ttl=63 time=96.8 ms
64 bytes from 10.32.161.105: icmp_seq=10 ttl=63 time=96.8 ms (DUP!)
64 bytes from 10.32.161.105: icmp_seq=11 ttl=63 time=122 ms
64 bytes from 10.32.161.105: icmp_seq=11 ttl=63 time=122 ms (DUP!)
64 bytes from 10.32.161.105: icmp_seq=11 ttl=63 time=122 ms (DUP!)
64 bytes from 10.32.161.105: icmp_seq=12 ttl=63 time=16.8 ms
64 bytes from 10.32.161.105: icmp_seq=13 ttl=63 time=62.4 ms
64 bytes from 10.32.161.105: icmp_seq=14 ttl=63 time=41.7 ms
64 bytes from 10.32.161.105: icmp_seq=15 ttl=63 time=21.5 ms
64 bytes from 10.32.161.105: icmp_seq=16 ttl=63 time=53.8 ms
64 bytes from 10.32.161.105: icmp_seq=17 ttl=63 time=226 ms
64 bytes from 10.32.161.105: icmp_seq=18 ttl=63 time=15.6 ms
64 bytes from 10.32.161.105: icmp_seq=19 ttl=63 time=189 ms
64 bytes from 10.32.161.105: icmp_seq=20 ttl=63 time=314 ms
64 bytes from 10.32.161.105: icmp_seq=21 ttl=63 time=12.8 ms

```

```

64 bytes from 10.32.161.105: icmp_seq=51 ttl=63 time=9.94 ms
64 bytes from 10.32.161.105: icmp_seq=52 ttl=63 time=58.1 ms
64 bytes from 10.32.161.105: icmp_seq=53 ttl=63 time=105 ms
64 bytes from 10.32.161.105: icmp_seq=54 ttl=63 time=13.0 ms
64 bytes from 10.32.161.105: icmp_seq=55 ttl=63 time=12.1 ms
64 bytes from 10.32.161.105: icmp_seq=56 ttl=63 time=123 ms
64 bytes from 10.32.161.105: icmp_seq=57 ttl=63 time=9.01 ms
64 bytes from 10.32.161.105: icmp_seq=58 ttl=63 time=72.3 ms
64 bytes from 10.32.161.105: icmp_seq=59 ttl=63 time=18.4 ms
64 bytes from 10.32.161.105: icmp_seq=60 ttl=63 time=11.5 ms
64 bytes from 10.32.161.105: icmp_seq=61 ttl=63 time=237 ms
64 bytes from 10.32.161.105: icmp_seq=62 ttl=63 time=10.9 ms
64 bytes from 10.32.161.105: icmp_seq=63 ttl=63 time=205 ms
64 bytes from 10.32.161.105: icmp_seq=64 ttl=63 time=12.7 ms
64 bytes from 10.32.161.105: icmp_seq=65 ttl=63 time=20.2 ms
64 bytes from 10.32.161.105: icmp_seq=66 ttl=63 time=251 ms
64 bytes from 10.32.161.105: icmp_seq=67 ttl=63 time=78.9 ms
64 bytes from 10.32.161.105: icmp_seq=68 ttl=63 time=100 ms
64 bytes from 10.32.161.105: icmp_seq=69 ttl=63 time=20.3 ms
64 bytes from 10.32.161.105: icmp_seq=70 ttl=63 time=139 ms
64 bytes from 10.32.161.105: icmp_seq=71 ttl=63 time=67.1 ms
64 bytes from 10.32.161.105: icmp_seq=72 ttl=63 time=9.04 ms
64 bytes from 10.32.161.105: icmp_seq=73 ttl=63 time=9.11 ms
64 bytes from 10.32.161.105: icmp_seq=74 ttl=63 time=24.8 ms
64 bytes from 10.32.161.105: icmp_seq=75 ttl=63 time=15.0 ms
64 bytes from 10.32.161.105: icmp_seq=76 ttl=63 time=14.0 ms
64 bytes from 10.32.161.105: icmp_seq=77 ttl=63 time=11.7 ms
64 bytes from 10.32.161.105: icmp_seq=78 ttl=63 time=208 ms
64 bytes from 10.32.161.105: icmp_seq=79 ttl=63 time=28.9 ms

```

Data Node 2

```
samyuktha_new@samyukthanew:~$ pwd
```

```
/home/samyuktha_new
```

```
samyuktha_new@samyukthanew:~$ echo $SPARK_HOME
```



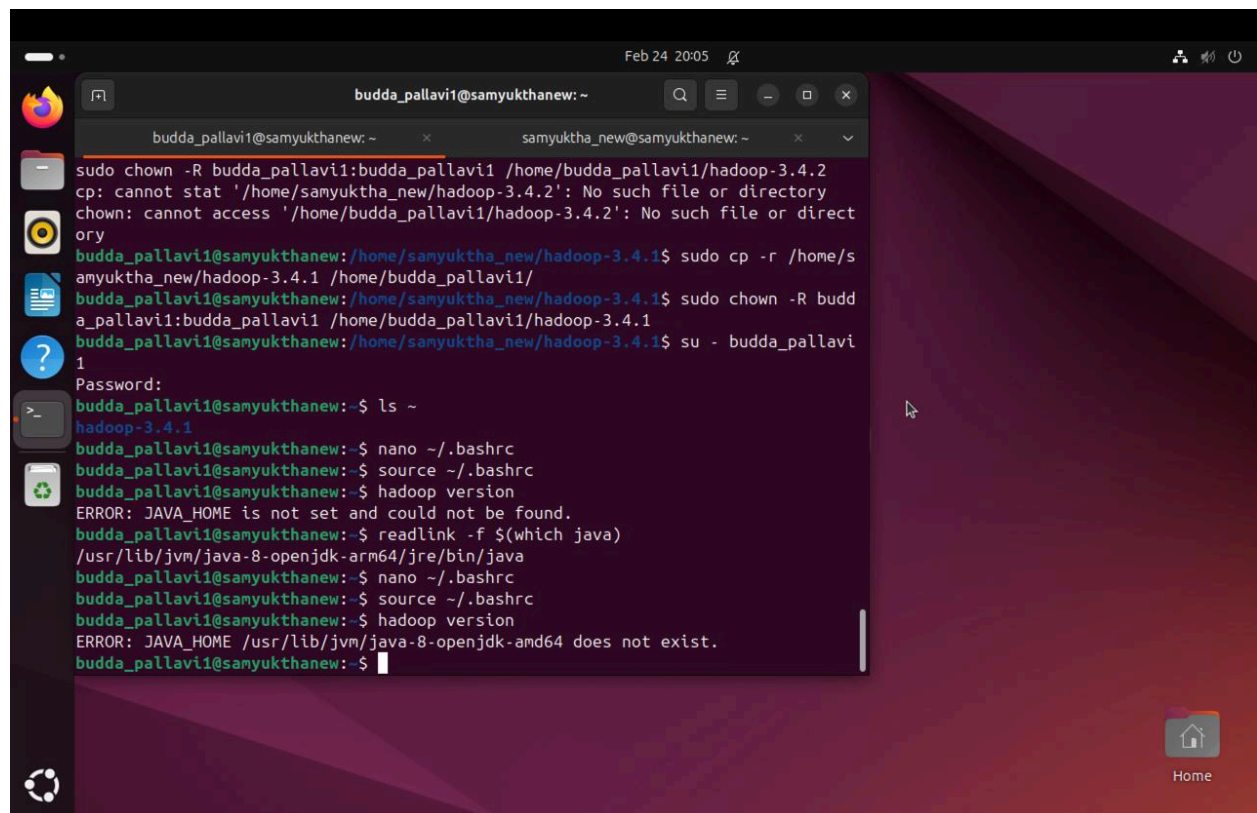
```

/home/samyuktha_new/spark
samyuktha_new@samyukthanew:~$ ls $SPARK_HOME/bin
beeline                pyspark                spark-class.cmd        spark-shell.cmd
beeline.cmd            pyspark2.cmd          spark-connect-shell     spark-sql
docker-image-tool.sh   pyspark.cmd           sparkR                  spark-sql2.cmd
find-spark-home        run-example            sparkR2.cmd            spark-sql.cmd
find-spark-home.cmd    run-example.cmd        sparkR.cmd             spark-submit
load-spark-env.cmd     spark-class            spark-shell             spark-submit2.cmd
load-spark-env.sh      spark-class2.cmd       spark-shell2.cmd        spark-submit.cmd
samyuktha_new@samyukthanew:~$ spark-submit --version
Welcome to

```

```
Using Scala version 2.12.18, OpenJDK 64-Bit Server VM, 1.8.0_482
Branch HEAD
Compiled by user runner on 2026-01-12T04:16:44Z
Revision 5a48a37b2dbd7b51e3640cd1d947438459556cc6
Url https://github.com/apache/spark
Type --help for more information.
samyuktha_new@samyukthanew:~$ cd ~/spark
sbin/stop-worker.sh
sbin/start-worker.sh spark://10.32.161.105:7077
stopping org.apache.spark.deploy.worker.Worker
starting org.apache.spark.deploy.worker.Worker, logging to
/home/samyuktha_new/spark/logs/spark-samyuktha_new-org.apache.spark.deploy.worke
r.Worker-1-samyukthanew.out
samyuktha_new@samyukthanew:~/spark$ jps
6803 Jps
6744 Worker
samyuktha_new@samyukthanew:~/spark$ grep -i "registered"
~/spark/logs/spark*Worker*.out
samyuktha_new@samyukthanew:~/spark$
```

```
a_new@samyukthanew:~/spark$ ping 10.32.161.105
PING 10.32.161.105 (10.32.161.105) 56(84) bytes of data.
64 bytes from 10.32.161.105: icmp_seq=1 ttl=64 time=100 ms
64 bytes from 10.32.161.105: icmp_seq=2 ttl=64 time=15.7 ms
64 bytes from 10.32.161.105: icmp_seq=3 ttl=64 time=149 ms
64 bytes from 10.32.161.105: icmp_seq=4 ttl=64 time=803 ms
64 bytes from 10.32.161.105: icmp_seq=5 ttl=64 time=110 ms
64 bytes from 10.32.161.105: icmp_seq=6 ttl=64 time=105 ms
64 bytes from 10.32.161.105: icmp_seq=7 ttl=64 time=75.4 ms
64 bytes from 10.32.161.105: icmp_seq=8 ttl=64 time=8.60 ms
64 bytes from 10.32.161.105: icmp_seq=9 ttl=64 time=22.9 ms
^C
--- 10.32.161.105 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8023ms
rtt min/avg/max/mdev = 8.595/154.348/802.986/233.829 ms
samyuktha_new@samyukthanew:~/spark$
```



```
Feb 24 19:59
budda_pallavi1@samyukthanew: /home/samyuktha_new/hadoop-3.4.1
budda_pallavi1@samyukthanew: /home/samy... x samyuktha_new@samyukthanew: ~
samyuktha_new@samyukthanew:~$ ls
amazon_q2          input.txt
amazon_q2_spark_bilstm.py  Music
dbpedia_lab4       Pictures
Desktop            Public
Documents           rdd_lab_final1.py
Downloads           rdd_lab_final2.py
employee.txt        rdd_lab_final.py
hadoop-3.4.1        sample.txt
hadoop-3.4.1.tar.gz searchterms.csv
hadoop_wordcount    snap
indicorp_q1_analysis1.py  spark
indicorp_q1_analysis_final.py  spark-3.5.8-bin-hadoop3.tgz
indicorp_q1_analysis.py    Templates
input1.txt               Videos

samyuktha_new@samyukthanew:~$ cd hadoop-3.4.1
samyuktha_new@samyukthanew:~/hadoop-3.4.1$ sudo adduser budda_pallavi1
fatal: The user 'budda_pallavi1' already exists.
samyuktha_new@samyukthanew:~/hadoop-3.4.1$ su budda_pallavi1
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

budda_pallavi1@samyukthanew: /home/samyuktha_new/hadoop-3.4.1$
```

```
Feb 24 19:48
samyuktha_new@samyukthanew: ~/spark
samyuktha_new@samyukthanew:~/spark$ jps
3372 Worker
3452 Jps
samyuktha_new@samyukthanew:~/spark$ whoami
samyuktha_new
samyuktha_new@samyukthanew:~/spark$ sudo nano /etc/ssh/sshd_config
[sudo] password for samyuktha_new:
Sorry, try again.
[sudo] password for samyuktha_new:
samyuktha_new@samyukthanew:~/spark$ sudo systemctl restart ssh
samyuktha_new@samyukthanew:~/spark$ whoami
samyuktha_new
samyuktha_new@samyukthanew:~/spark$ sudo adduser budda_pallavi1
sudo usermod -aG sudo budda_pallavi1
info: Adding user 'budda_pallavi1' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'budda_pallavi1' (1002) ...
info: Adding new user 'budda_pallavi1' (1002) with group 'budda_pallavi1 (1002)'
...
info: Creating home directory '/home/budda_pallavi1' ...
info: Copying files from '/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
```



```

[hadoop@master:~]$ hdfs dfs -ls /output/q1
2026-02-24 16:38:10,541 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 6 items
drwxr-xr-x - hadoop supergroup          0 2026-02-24 16:32 /output/q1/task1_stats
drwxr-xr-x - hadoop supergroup          0 2026-02-24 16:32 /output/q1/task2_sentence_length
drwxr-xr-x - hadoop supergroup          0 2026-02-24 16:32 /output/q1/task3_tokens_assamese
drwxr-xr-x - hadoop supergroup          0 2026-02-24 16:33 /output/q1/task3_tokens_odia
drwxr-xr-x - hadoop supergroup          0 2026-02-24 16:35 /output/q1/task3_tokens_urdu
drwxr-xr-x - hadoop supergroup          0 2026-02-24 16:36 /output/q1/task4_tfidf_top20_as
[hadoop@master:~]$ hdfs dfs -cat /output/q1/task1_stats/part*
2026-02-24 16:38:20,662 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
3375484,1687742
[hadoop@master:~]$ hdfs dfs -cat /output/q1/task2_sentence_length/part*
2026-02-24 16:38:28,129 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
3,97453,261.45550445506484
[hadoop@master:~]$ hdfs dfs -cat /output/q1/task3_tokens_assamese/part*
2026-02-24 16:38:39,587 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
অক্ষ,1021426
এই,563871
কৰা,538699
পৰা,516678
ক'ব,497801
বা'ব,393470
ইহা,310129
বুলি,275982
কৰা'ব,230236
(য,217215
ক'ব'ব,205295
।,196416
লাগ'ত,191490
ক'ব।,184377
এক,182758
।,182253
ইহা'হা,181943
ক'ব,180906
(অক্ষ,179253
(হা'হা,176838
ক'ব',169146
এটা,158555
ক'ব'ব'ল,148248
ইল,145844
অক্ষ,133426
কিছু,124758
(সেই,121304
বিজ্ঞান,120395
বা,120212
(ক'না,117531

```

```

[hadoop@master:~]$ hdfs dfs -cat /output/q1/task3_tokens_odia/part*
2026-02-24 16:38:47,800 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
।,4001922
ও,1320538
ଏହି,1042956
ପା'ର,957939
ବୋ,631718
ଏବଂ,631578
ଏବଂ,601491
ମଧ୍ୟ,537399
ପରେ,531047
କିଛି,487445
କିଛି,428233
ବୋ'ଲ,424977
ବୋ'ଲ,393031
କିଛି,384385
ଏହା,366484
ବୋ'ଲ,333870
ପା'ର,292915
ବୋ'ଲ,283673
ମଧ୍ୟ,283029
କିଛି,270814
କିଛି,260701
କିଛି,258950
କିଛି,247916
ମଧ୍ୟ,244777
କିଛି,242346
କିଛି,226982
କିଛି,225962
କିଛି,225169
କିଛି,221092
କିଛି,220840

```

```

hadoopmaster:~$ hdfs dfs -cat /output/q1/task3_tokens_urdu/part*
2026-02-24 16:39:00,175 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
24851366,
18121582,
18849552,
13882690,
12915737,
10082299,
9298421,
9065252,
8879204,
8088289,
7687252,
6797519,
4599471,
4442363,
4118193,
4027396,
4003359,
3527988,
3433591,
3374679,
3358429,
3175707,
3129778,
, 2580097
2428175,
2424906,
2228247,
2281752,
2002528,
1953514,
hadoopmaster:~$ hdfs dfs -cat /output/q1/task4_tfidf_top20_as/part*
2026-02-24 16:39:22,430 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
", 1949585
, 1021426
, 563871
, 538699
, 516678
, 497881
, 393470
, 310129
, 275982
, 230236
, 217215
, 205295
, 196416
, 191490
, 184377
, 182758
, 182253
, 181943
, 180986
, 179253

```

```

2026-02-24 17:10:00 INFO PathNotFoundException: PathNotFoundException
>>> Loading data from HDFS
>>> Cleaning and tokenizing text
>>> Building vocabulary
>>> Collecting data to driver
>>> Training started
Epoch 1/2 completed
Epoch 2/2 completed
Accuracy on CLEAN TEST DATA: 84.16%
Accuracy on NOISY TEST DATA: 80.40%

```